

UX / PRODUCT DESIGN CASE STUDY

# Reimagining Saved Places

A System Redesign for Saving, Organizing, and Acting on Locations in Google Maps

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**Product:** Google Maps — Saved Places & Lists

**Scope:** Smart Collections, Trip Planning, Collaborative Itineraries, Contextual Categorization

**Research base:** Travelers, Students, Commuters, Delivery Workers

# 1. Problem Statement & Context

## 1.1 Framing the Problem

Google Maps treats a saved place as a single, flat unit: a pin with a label and an optional list assignment. This model has not changed materially since list-based saving launched, while user behavior around saving has grown considerably more complex. People now save places for reasons that span time horizons, group membership, and operational context, and the product gives them one structure to express all of it: a static, undifferentiated list.

The result is accumulation without organization. Saved Places grows linearly with use and degrades in usefulness at the same rate, because the system provides no mechanism to differentiate a place saved for tomorrow's coffee run from one saved for a trip planned eight months out.

### Problem Statement (Lean UX format)

We believe that **frequent Google Maps users who save 10 or more places** experience reduced trust and slower retrieval **because the saving system offers no automated structure, temporal context, or shared access.**

We will know this is true when we see **a reduction in median time-to-navigate from a saved item, an increase in 30-day retention on multi-person lists, and a measurable drop in users exporting trip plans to third-party tools.**

- How might we let Saved Places organize itself around how, when, and with whom a place will actually be used?
- How might we reduce the manual labor of list curation without removing user control over categorization?
- How might we make a saved list collaborative and chronologically aware without turning Google Maps into a project management tool?

## 1.2 Business Impact

Saved Places sits inside the core retention loop of Google Maps: a user saves a place, returns to Maps to retrieve it, and that return visit reinforces Maps as the default spatial decision-making surface. When the saved-places experience degrades, that loop breaks at the retrieval step, and the user substitutes another tool for the second half of the journey.

- Drop-off at retrieval: Users with large, unsorted save lists take measurably longer to find a specific place, which increases abandonment of the saved-places surface in favor of re-searching from scratch.
- Leakage to third-party planning tools: Multi-stop trip planning, especially in groups, has no native home in Maps. This pushes the actual planning work into Notion, Google Sheets, or shared documents, and Maps is relegated to single-destination navigation rather than the planning surface of record.

- **Reduced session frequency:** Each trip planned outside Maps represents a set of return visits to Maps that never happen organically; the user opens Maps only to pull a single address rather than to browse, compare, and decide.
- **Weakened data signal:** Poorly structured saves degrade the quality of behavioral data Google can use to power recommendations, since the system cannot distinguish an aspirational save from an active plan.

## 1.3 User Impact

For the end user, the cost is cognitive rather than financial. Every additional unsorted save increases the scanning effort required to retrieve the right one, and every list that mixes purposes — a birthday restaurant next to a delivery drop-off next to a dream vacation site — adds a small tax to an action that should be near-instant.

- **High cognitive load at retrieval,** particularly for users with 50+ saved places spread across few or no lists.
- **Loss of temporal context:** a save carries no inherent sense of when it is relevant, so old and current intentions sit side by side indefinitely.
- **No native mechanism for shared decision-making,** forcing groups to coordinate plans through screenshots, copied links, and external documents.
- **Visual clutter on the map canvas itself,** where dense pin clusters from unrelated save categories obscure the items a user is actually looking for in the moment.

## 2. User Research & Insights

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Research was structured around four user groups selected for behavioral distinctiveness rather than demographic variety: travelers, students, commuters, and delivery workers. Each group interacts with the save-and-retrieve loop at a different frequency, with different stakes attached to getting organization wrong.

### 2.1 Behavioral Synthesis by Group

#### Travelers

Travelers think in sequence and in groups. A save is rarely a standalone decision; it is one stop in a larger arc that includes a flight, a hotel, and several other travelers who need visibility into the same plan. Their core friction is that Maps captures the place but discards the plan: once five restaurants and three landmarks are saved for a single trip, nothing in the product encodes which day each belongs to, or who else is going.

This pushes travelers toward parallel tools almost immediately after the saving phase. The save becomes raw material that gets manually re-typed into a spreadsheet or document the moment more than one person needs to coordinate around it.

#### Students

Students organize around constraint, primarily budget and proximity to campus or home, rather than around time. Their saves cluster geographically and economically: cheap eats near campus, late-night study spots, affordable grocery options within walking distance. The defining behavior is repeated filtering by price tier and distance, done manually because Maps has no concept of a budget-aware cluster.

Because students revisit the same small geographic radius daily, their frustration is less about losing track of saves and more about the list never adapting; a place that closed or became unaffordable lingers in the list indefinitely, and new discoveries get added without ever being reconciled against what is already there.

#### Commuters

Commuters operate on a fixed, repeated route and save places almost incidentally: a pharmacy slightly off the usual path, a coffee shop near a transfer point. Their organizational need is the smallest of the four groups but the most route-dependent. They want micro-conveniences surfaced only when relevant to the current commute, not buried in a generic list they rarely open.

The friction here is invisibility, not clutter: useful saves exist but never resurface at the moment they would matter, so commuters frequently re-search for the same nearby pharmacy or coffee shop rather than trusting that it is already saved somewhere.

#### Delivery Workers

Delivery users have the highest save frequency and the lowest tolerance for friction. Their saves are transactional and high-volume: drop-off points, building entrances, gate codes, and parking spots

accumulate across a single shift. Sequencing matters more than naming or categorization; the operational question is always which stop is next, in what order, not what kind of place this is.

Generic list structures actively work against this group. A flat, alphabetically or recency-sorted list forces manual mental re-sequencing on every shift, and there is no way to mark a save as expired once a delivery route is complete, so the list accumulates dead entries at a faster rate than any other group.

## 2.2 Comparative Research Summary

| User Group       | Save Frequency                                 | Primary Frustration                                            | Organization Goal                                       |
|------------------|------------------------------------------------|----------------------------------------------------------------|---------------------------------------------------------|
| Travelers        | Bursty — high volume in short pre-trip windows | No chronological or multi-person structure to saved trip stops | Sequenced, shareable itinerary across days              |
| Students         | Moderate — steady, recurring additions         | Lists do not filter by budget or proximity automatically       | Budget- and distance-aware local clusters               |
| Commuters        | Low — occasional, route-incidental saves       | Useful saves never resurface at the relevant moment            | Context-triggered micro-suggestions along a fixed route |
| Delivery Workers | Very high — dozens per shift                   | No sequencing logic; manual re-ordering every shift            | Auto-sequenced, high-density, expiring drop-off lists   |

*A consistent pattern emerges across all four groups despite their different stakes: every group eventually performs a manual sorting task that the system should be able to infer from existing signals — date of intended use, location density, price tier, or route proximity. The product currently asks the user to provide structure manually that it already has enough data to generate automatically.*

## 3. Personas & Pain Points

### 3.1 Persona One: The Multi-Stop Leisure Traveler

| Attribute        | Detail                                                                                                                                     |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Name             | Priya Subramaniam                                                                                                                          |
| Context          | Plans 2–3 multi-day trips per year, usually with a group of 3–5 friends or family                                                          |
| Behavioral trait | Saves places in bursts during late-night research sessions, often 15–20 in one sitting, across multiple cities for the same trip           |
| Behavioral trait | Re-types saved places into a shared Google Doc within 48 hours of saving them, because the group needs to see and comment on the same list |
| Behavioral trait | Checks the map the morning of an activity to confirm logistics, then never returns to that saved place again                               |
| Core need        | A way to see saves as a sequence tied to specific days and shared with named collaborators, not as an undifferentiated pin pile            |

### 3.2 Persona Two: The High-Frequency Delivery Agent

| Attribute        | Detail                                                                                                                                                      |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Name             | Marcus Webb                                                                                                                                                 |
| Context          | Completes 25–40 deliveries per shift across a rotating set of neighborhoods                                                                                 |
| Behavioral trait | Saves a new drop-off detail (gate code, buzzer number, preferred parking spot) almost every shift, rarely revisiting the same exact address twice in a week |
| Behavioral trait | Manually reorders his mental route each shift because the app provides no route-aware sequencing of saved stops                                             |
| Behavioral trait | Never deletes old saves because doing so takes more time than leaving them, so his list has grown to several hundred entries with no expiration             |
| Core need        | Auto-sequenced, route-optimized groupings of active stops with saves that expire or archive themselves once a shift ends                                    |

### 3.3 Core Pain Points

#### 1. List Fragmentation

Both personas generate saves faster than they can organize them. Priya's trip research produces a burst of saves with no automatic grouping by trip or destination city; Marcus's shift produces dozens of saves with no automatic grouping by route or shift date. In both cases, the burden of grouping falls entirely on the user, at the exact moment they have the least time to do it.

## **2. Lack of Temporal and Operational Context**

Neither persona's saves carry any sense of when they stop being relevant. Priya's post-trip saves linger alongside next year's pre-trip research. Marcus's completed-shift drop-offs linger alongside tomorrow's active stops. The system has no concept of a save's lifecycle, so the list only ever grows.

## **3. No Native Multi-User Syncing**

Priya's group needs to view, comment on, and reorder the same itinerary in real time, but Maps offers no shared editing surface, forcing the plan to live in a separate document the moment more than one person is involved. This severs the connection between the planning artifact and the navigation tool that will ultimately be used to execute the plan.

## **4. Visual Clutter on the Viewport**

Marcus's several hundred undifferentiated saves render as a dense, undifferentiated pin field whenever his saved layer is visible, making it difficult to isolate the handful of stops relevant to the current shift. Priya experiences the same problem at smaller scale: pins from a prior trip remain visually indistinguishable from pins for the trip she is currently planning.

## 4. User Flows & Wireframe Descriptions

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### 4.1 Flow One: Creating a Collaborative Multi-Stop Weekend Itinerary

This flow follows Priya from an existing set of unsorted saves to a shared, day-structured itinerary that her group can co-edit.

#### Critical Path

1. From the Saved tab, the user selects multiple existing saved places via a long-press multi-select interaction, or starts a new collection directly from search results.
2. The system prompts: "Turn these into a trip?" with a single tap to confirm, triggering creation of a new Itinerary object distinct from a standard list.
3. The user assigns trip dates (e.g., a 3-day range); the system auto-distributes saved places across days using proximity clustering as a starting suggestion, fully overridable by drag-and-drop.
4. The user taps "Invite" to add collaborators by phone number, email, or shareable link; invited members receive edit or view-only permissions, set per person.
5. Any collaborator can reorder stops within a day, add a new stop directly from their own device, or leave a lightweight comment on a stop; changes sync in real time across all participants.
6. On the day of travel, the itinerary surfaces as a persistent bottom sheet with the current day's sequence pre-loaded and one-tap navigation to the next stop.

#### Wireframe Specification — Itinerary Builder Screen

- Top app bar: trip title (editable inline), date range pill, and an avatar stack showing collaborators with a "+" affordance to invite more.
- Primary layout: persistent contextual bottom sheet using a split-view structure — a vertical day-by-day timeline occupies the left two-thirds of the sheet at full height, while a live map preview occupies the right third, updating to highlight the currently selected day's pins.
- Timeline column: each day renders as a collapsible section header ("Day 1 — Sat, Oct 4") containing draggable stop cards; each card shows place name, a small thumbnail, estimated travel time to the next stop, and an overflow menu for remove/comment/reassign-day.
- Day reassignment: drag-and-drop between day sections, with a drop-target highlight using the accent color and inertial scroll when dragging near the top or bottom edge of the visible area.
- Pill filters above the timeline: "All," "Food," "Activities," "Lodging" — tapping a pill filters the visible cards without altering the underlying itinerary data.
- Progressive disclosure: collaborator comments are collapsed by default behind a small comment-count badge on each card, expanding inline only when tapped, to avoid timeline clutter.
- Bottom sheet handle supports three states: collapsed (shows only current/next stop as a single row), half-height (shows the active day), and full-height (shows the complete multi-day timeline).



## 4.2 Flow Two: Converting a Messy List into an Automated Smart Collection

This flow follows Marcus as he transforms an unsorted accumulation of saves into a self-maintaining collection filtered by proximity and operating hours.

### Critical Path

7. From an existing flat list, the user taps "Organize with Smart Collections," surfaced as a suggestion banner once a list exceeds a save-count threshold.
8. The system analyzes the list's existing places and proposes 2–4 candidate Smart Collections (e.g., "Open Now Near You," "Downtown Cluster," "This Week's Route"), each shown as a preview card with a sample of the places it would contain.
9. The user selects a proposed collection or customizes the filter logic directly, choosing from condition pills: proximity radius, operating-hours window, last-saved date, or custom tag.
10. On confirmation, the Smart Collection is created as a live, rule-based view — it is not a static copy of places, but a filter that re-evaluates membership automatically as the user's location, time, or place data changes.
11. Places that no longer match the rule (e.g., a location now closed, or outside today's route radius) automatically move to an "Archived" sub-state rather than disappearing, preserving recoverability.
12. The user can pin a Smart Collection to the top of the Saved tab and to the map's quick-access layer for one-tap visibility during active use.

### Wireframe Specification — Smart Collection Configuration Screen

- Entry banner: appears inline at the top of a flat list once it crosses a save-count threshold, using a light accent background, a short headline ("This list is getting long — want Maps to organize it?"), and a single primary action button.
- Canvas layering: the configuration screen layers a translucent rule-builder panel over a dimmed, partially visible map canvas beneath it, so the user retains spatial context for the places being filtered while focusing on rule logic in the foreground layer.
- Rule-builder panel: a vertically stacked set of condition rows, each row pairing a labeled pill filter (e.g., "Within \_\_\_ miles") with an inline control — a slider for radius, a time-range picker for hours, a toggle for tag matching.
- Live preview strip: directly beneath the rule-builder, a horizontally scrollable row of place cards updates in real time as rules change, giving immediate visual feedback on which saved places currently qualify.
- Footer action bar: a sticky two-button row — "Save as Smart Collection" (primary, accent-filled) and "Save as static list instead" (secondary, outlined) — ensuring users retain the option to opt out of automation.
- Density toggle: a compact/comfortable view switch in the top-right of the preview strip, addressing the high-volume use case where dozens of small cards need to be scannable without excessive scrolling.

## 5. Final Recommendations & Success Metrics

### 5.1 Product Recommendations

- Introduce a new object type, the Itinerary, as a sibling to the existing List object within the Saved Places data model, rather than overloading Lists with date and collaborator fields they were not designed to hold.
- Implement Smart Collections as rule-based saved views rather than static snapshots, so that proximity, operating-hours, and recency filters re-evaluate membership automatically as underlying place and location data changes.
- Add a lightweight permissions model to collaborative itineraries — owner, editor, viewer — reusing the sharing and identity infrastructure already in place for Google Workspace and Google Photos sharing, rather than building a parallel system.
- Surface organizational prompts contextually and sparingly: trigger the "organize this list" banner only past a save-count threshold, to avoid interrupting low-frequency users like commuters who do not need the feature.
- Default new Smart Collections and Itineraries to a soft-archive behavior rather than deletion, since irreversible data loss is the primary reason users distrust automated organization features.
- Phase the rollout: ship contextual categorization and Smart Collections first, since they require no real-time infrastructure; ship collaborative itineraries in a second phase once multiplayer sync is validated at scale.

### 5.2 Success Metrics

| Metric                                   | Type     | Target Signal                                                                                    |
|------------------------------------------|----------|--------------------------------------------------------------------------------------------------|
| Task success rate for itinerary creation | UX       | Percentage of users who complete a multi-stop itinerary without abandoning the flow              |
| 30-day retention on collaborative lists  | Business | Share of created itineraries with at least one collaborator action 30 days post-creation         |
| Time-to-navigate from a saved item       | UX       | Median seconds from opening Saved Places to tapping "Directions" on a specific place             |
| Churn to external planning tools         | Business | Self-reported and inferred (e.g., copy-link events) drop in trip planning conducted outside Maps |
| Smart Collection adoption rate           | Product  | Percentage of eligible users (lists above threshold) who accept a Smart Collection suggestion    |
| Stale save rate                          | UX       | Share of saved places never revisited or actioned within 90 days, tracked pre/post launch        |

### 5.3 Technical Constraints & Engineering Hand-Off Considerations

- Real-time multiplayer syncing for collaborative itineraries requires a conflict-resolution strategy for simultaneous edits (e.g., two collaborators reordering the same day); a last-write-wins model is insufficient and should be evaluated against operational transform or CRDT-based approaches already used elsewhere at Google.
- Smart Collections depend on live place metadata (hours, status, closures); rule evaluation must handle stale or missing data gracefully, with a defined fallback when a place's operating-hours data is unavailable rather than silently excluding it.
- Offline access is a hard requirement for the delivery-worker segment specifically; itineraries and Smart Collections must cache their last-resolved state locally and clearly indicate when a view is showing potentially outdated data due to lack of connectivity.
- Permissions and sharing should be built on existing Google identity and sharing infrastructure to avoid duplicating access-control logic, but this introduces a cross-team dependency that should be scoped early given the phased rollout plan.
- The proximity- and route-based clustering used in both Smart Collections and itinerary day-distribution should reuse existing Maps routing and ETA infrastructure rather than introducing a parallel geospatial clustering service, to limit latency and maintenance overhead.

### 5.4 Conclusion

This redesign reimagines Google Maps Saved Places as an intelligent planning tool rather than a static bookmarking feature. By introducing Smart Collections and Collaborative Itineraries, the proposed solution reduces manual organization, improves retrieval, and supports real-world planning and collaboration, creating a more seamless and engaging user experience.